Pressure Metrics and Forward ATS Performance

NFL regular season 2015 to 2025. Source: nflfastR play-by-play (nflverse). Generated 2026-10-03.

Bottom line

Pressure has signal, but it is role-dependent. The same pressure profile covers as an underdog and fails as a favorite. The one clean directional read is sacks: defenses that cannot get home do not cover, and offenses that keep sacks off the sheet do. None of it is as strong as the defensive-EPA regression signal from the broader screen.

Method

For each season and team, pressure metrics were computed over the first K games (K = 3, 4, 5, 6). Teams were ranked within season into quintiles. ATS results were then measured in games K+1 and K+2. Spread is the closing line from the nflverse schedule file. Pushes are excluded from cover percentages. Favorite and underdog splits use the sign of the spread. Results below are pooled across the four windows unless a window is named. Each window covers 352 team-seasons and roughly 700 team-games.

Definitions (per dropback, where dropback = pass attempts + sacks):

- **Pressure allowed** = (QB hits taken + sacks taken) / offensive dropbacks
- **Pressure generated** = (QB hits + sacks) / defensive dropbacks faced
- **Sack rate allowed / generated** = sacks / dropbacks, each side
- **Sack conversion** = sacks / (QB hits + sacks). Offensive side measures how often a pressure allowed became a sack. Defensive side measures how often a pressure generated finished.
- **Sack differential** = sack rate generated minus sack rate allowed
- QB hit is the nflfastR qb_hit flag. No charting-based pressure data is used.

1. Pressure allowed

Quintile (through K games)	Pooled ATS	Cover %	As favorite	Fav %	As dog	Dog %
Most pressured offenses	299-295	50.3	93-127	42.3	206-168	55.1
Least pressured offenses	260-250	51.0	142-139	50.5	118-109	52.0

A leaky offensive line is only a fade when the market still has that team laying points. The most-pressured quintile is below 50% as a favorite in 3 of 4 windows (K=3: 21-40, K=4: 20-35, K=5: 23-22, K=6: 29-30). As an underdog it is above 50% in all 4 windows.

2. Pressure generated (QB hit rate)

Quintile (through K games)	Pooled ATS	Cover %	As favorite	Fav %	As dog	Dog %
Most hits generated	299-303	49.7	148-193	43.4	151-110	57.9
Fewest hits generated	260-254	50.6	108-102	51.4	152-152	50.0

Same shape as pressure allowed, stronger. Pressure-heavy defenses laying points go 30-50 in both the K=4 and K=5 windows. As underdogs they cover at 57.9% on 261 games. Pressure defenses play up to the opponent and down to the spread.

3. Sacks: the directional signal

Feature	Quintile	Pooled ATS	Cover %	As favorite	Fav %	Windows > 50%
Sack rate generated	Fewest	234-279	45.6	100-137	42.2	0 of 4

Feature	Quintile	Pooled ATS	Cover %	As favorite	Fav %	Windows > 50%
Sack rate generated	Most	292-306	48.8	172-179	49.0	2 of 4
Sack rate allowed	**Fewest**	**272-243**	**52.8**	**177-151**	**54.0**	**3 of 4**
Sack rate allowed	Most	290-300	49.2	94-116	44.8	1 of 4
Sack differential	Best	313-291	51.8	212-185	53.4	3 of 4
Sack differential	Worst	242-265	47.7	83-101	45.1	1 of 4

Fewest sacks generated is the cleanest fade in the set: 44.2%, 43.7%, 45.3%, 43.7% by window. It is never above 50%. Fewest sacks allowed is a back, and the K=6 bucket is 77-52 (59.7%), 7 of 11 seasons above 50%.

4. Sack conversion (share of pressures that end in a sack)

Side	Quintile	Pooled ATS	Cover %	As favorite	Fav %	Windows > 50%
Offense (pressures allowed)	**Lowest conversion**	**269-242**	**52.6**	**181-145**	**55.5**	**3 of 4**
Offense (pressures allowed)	Highest conversion	291-298	49.4	124-140	47.0	1 of 4
Defense (pressures generated)	**Lowest conversion**	**234-272**	**46.2**	**105-135**	**43.8**	**0 of 4**
Defense (pressures generated)	Highest conversion	286-308	48.1	158-162	49.4	1 of 4

Offenses that avoid sacks when pressured cover, and do so as favorites (55.5%). That is a quarterback trait and it persists. Defenses that hit the quarterback but do not finish are a fade in every window. Hits without sacks do not translate.

5. Window-by-window detail

Cover percentage of the named quintile in games K+1 and K+2. Parenthetical is seasons above 50% out of 11.

Feature / quintile	K=3	K=4	K=5	K=6
Pressure allowed, most, as favorite	21-40	20-35	23-22	29-30
Pressure allowed, most, as dog	48-42	55-40	59-43	44-43
QB hit rate gen, most, as favorite	42-43	30-50	30-50	46-50
QB hit rate gen, most, as dog	37-30	43-27	39-29	32-24
Sack rate gen, fewest (all)	49.2 (3)	44.2 (2)	45.3 (2)	43.7 (3)
Sack rate allowed, fewest (all)	54.6 (5)	48.4 (4)	48.4 (4)	59.7 (7)
Sack conv off, lowest (all)	54.0 (6)	53.9 (7)	50.0 (5)	52.7 (5)
Sack conv def, lowest (all)	48.5 (4)	46.8 (2)	47.6 (4)	42.1 (2)
Sack differential, best, as favorite	51-43	47-46	55-46	59-50

6. Usable ranking

1. Fade defenses in the bottom quintile for sacks generated, especially as favorites.
2. Fade pressure-generating defenses as favorites. Back them as underdogs.
3. Fade heavily-pressured offenses as favorites. Back them as underdogs.
4. Back offenses that keep sacks off the sheet, as favorites.
5. Back offenses with the lowest pressure-to-sack conversion, as favorites.

Pressure differential by itself is noise. Raw QB hit rate allowed and most-sacks-generated carry no stable directional edge outside the favorite/underdog split.

Notes

Ten pressure features were screened across four windows. Buckets are roughly 125 to 150 team-games per window and 500 to 600 pooled. A favorite/underdog split that nets to 50% overall but diverges by role is a real deviation from baseline, because every game contributes one favorite and one underdog to the full sample.